Dr. Joshua L. Pughe-Sanford

youtube | email | website | scholar

Theoretical, computational, and mathematical physicist specializing in nonlinear dynamics and theories of learning and inference in neural systems

Experience

Research Scientist 2026–Present
Principia Princeton, NJ

Flatiron Research Fellow 2023–2026
Flatiron Institute, Center for Computational Neuroscience New York, NY

Education

Ph.D. in Physics · M.S. in Mathematics (GPA: 3.9) 2017–2023
Georgia Institute of Technology, Center for Non-Linear Dynamics Atlanta, GA

B.S. in Physics · Minor in Mathematics (GPA: 3.9, Highest Honors) 2013–2017
Emory University Atlanta, GA

Awards

Presidential Fellow · Herbert P. Haley Fellow · Phi Beta Kappa Honors Society

Teaching and Mentorship Experience

Flatiron Institute *2023–2026*

Mentored five graduate students, resulting in publication [1, 4].

Georgia Institute of Technology *2017–2023*

Mentored three undergraduate students, resulting in publication [6].

Promoted to lead teaching assistant, training and managing 30 graduate TAs.

Received on average **4.9/5** on teaching evaluations.

Taught graduate lectures on nonlinear dynamics and mathematical methods for physicists.

Service

Diversity, Equity, and Inclusion Committee, *at Simons Foundation* *2024–2026*

Advocated for more inclusive hiring practices across the institution.

Led intern selection for my research group from 500+ applicants, prioritizing candidates underrepresented in science.

Physics Allies for Wellness, *at Georgia Institute of Technology* *2020–2023*

Co-founded an organization providing support, resources, and action plans for those facing injustice and inequity in the physics department.

Graduate Association of Physicists, *at Georgia Institute of Technology* *2018–2020*

Publications † indicates co-first authorship

[1] I. Kalburge, **J. L. Pughe-Sanford**, and A. Williams, "Sequential regression in Recurrent Neural Networks," *submitted*.

[2] **J. L. Pughe-Sanford**†, A. Mah†, S. Harvey, and A. Williams, "Linear approximations to HMM filtering," *NeurIPS* (2026).

[3] F. Noah, **J. L. Pughe-Sanford**, and S. Grauer, "Degree-of-freedom and optimization dynamic effects on the observability of the Kuramoto-Sivashinsky model," *JCP* (2026).

[4] S. Qin, **J. L. Pughe-Sanford**, A. Genkin, P. G. Ozdil, P. Greengard, A. M. Sengupta, and D. Chklovskii, "A network of biologically inspired Rectified Spectral Units (ReSUs) learns hierarchical features without error backpropagation," *AAAI*, 2026.

[5] **J. L. Pughe-Sanford**†, X. Ding†, J. J. Moore†, C. Epstein, P. Greengard, A. M. Sengupta, and D. Chklovskii, "Neurons as Detectors of Coherent Sets in Sensory Dynamics," *NeurIPS*, 2025.

[6] **J. L. Pughe-Sanford**, S. Quinn, T. Balabanski, and R. O. Grigoriev, "Computing chaotic time-averages from few periodic or non-periodic orbits," *Chaos*, 2025.

[7] J. J. Moore, A. Genkin, M. Tournoy, **J. L. Pughe-Sanford**, R. R. de Ruyter van Steveninck, and D. B. Chklovskii, "The Neuron as a Direct Data-Driven Controller," *PNAS*, 2024.

[8] **J. L. Pughe-Sanford**, "On the Generalization of Shadowing to Fluid Turbulence: Practical Methods For Quantifying Dynamical Similarity," Ph.D. Dissertation, Georgia Institute of Technology, 2023.

[9] **J. L. Pughe-Sanford**†, C. J. Crowley†, W. Toler, R. O. Grigoriev, and M. F. Schatz, "Observing a dynamical skeleton of turbulence in Taylor–Couette flow experiments," *Philosophical Transactions of the Royal Society A*, 2023.

[10] **J. L. Pughe-Sanford**†, C. J. Crowley†, W. Toler, M. C. Krygier, R. O. Grigoriev, and M. F. Schatz, "Turbulence tracks recurrent solutions," *PNAS*, 2022.

[11] M. C. Krygier, **J. L. Pughe-Sanford**, and R. O. Grigoriev, "Exact coherent structures and shadowing in turbulent Taylor–Couette flow," *Journal of Fluid Mechanics*, 2021.

[12] S. Boettcher and **J. L. Pughe-Sanford**, "Renormalization of Discrete-Time Quantum Walks with non-Grover Coins," *Journal of Statistical Mechanics: Theory and Experiment*, 2017.

[13] **J. L. Pughe-Sanford**, "Properties of Quantum Walks within Various One-Dimensional Media," Honors Thesis, Emory University, 2017.

Recorded Talks

J. L. Pughe-Sanford, S. Quinn, T. Balabanski, and R. O. Grigoriev, "Computing Chaotic Time-Averages from a Small Number of Periodic Orbits." APS DFD, 2023

J. L. Pughe-Sanford, and R. O. Grigoriev, "Vortex Interactions: a Low-Dimensional Approach to the Inverse Cascade." APS DFD, 2022

J. L. Pughe-Sanford, and R. O. Grigoriev, "Heteroclinic Connections as Predictors of Extreme Events in Weakly Turbulent Flow." APS DFD, 2019

Research Summary

Principia, *with A. Saxe and J. Lee* *2026–Present*

Building theoretical model organisms of learning: architectures simple enough to study from first principles, yet sophisticated enough to explore phenomena seen in large models. Personas and emergent misalignment are of particular interest.

Flatiron Institute, *with A. Williams* *2025–2026*

Investigated how sufficient statistics are represented in recurrent neural architectures during sequential inference tasks [1].

Characterized which hidden Markov processes are easily filtered by linear RNNs, shedding light on where complicated architecture is needed and, importantly, where it is *not* [2].

Flatiron Institute, *with D. Chklovskii* *2023–2025*

Engineered interpretable, dynamical models of neural computation through the lens of observability [5] and controllability [7].

Georgia Institute of Technology, *with R. Grigoriev* *2017–2023*

Generalized the Axiom A framework of low-dimensional chaos to fluid turbulence [8]. In particular, we presented evidence that turbulence is well predicted over short time intervals by invariant sets [9–11]. We then showed that, over long time scales, ergodic averages can be computed accurate from even a meager amount of such sets [6].

Emory University, *with S. Boettcher* *2015–2017*

Derived scaling laws for the rate at which a quantum wavefunction will spread out over a self-similar lattice, and validated these results *in silico* [12].

Explored computational complexity advantages of quantum versus classical algorithms [13].

Princeton University, *with F. Calaprice* *2013–2017*

Helped design, fabricate, and deploy a dark matter detection experiment.

Automated and improved the purification process that removes radioactive contaminants from the detector, improving its sensitivity.

Oxford College, *with R. Conceicao* *2013–2015*

Analyzed the Collatz conjecture and its polynomial variant, producing asymptotic approximations for total stopping time.

Hobbies

Robotics · Video Game Development · Volleyball · Camping